

Application-Layer Protocols

Definition: An **application-layer protocol** specifies how processes on different end systems exchange messages. It defines message types, structure, meaning, and exchange rules.

Core Components

- **Message Types:** e.g., request and response.
- **Syntax:** message fields and format.
- **Semantics:** meaning of each field.
- **Exchange Rules:** when and how messages are sent.

Categories

- **Public Protocols:** Open standards like **HTTP** (RFC 7230).
- **Proprietary Protocols:** Used in private apps (e.g., Skype).

Examples

1. Web (HTTP)

- Components: HTML, browsers, servers, and HTTP.
- Defines format and sequence of browser-server communication.

2. Netflix (DASH)

- Components: media servers, client apps, billing systems.
- **DASH** protocol handles adaptive video streaming.

Note: A protocol is only one part of an application. For instance, HTTP enables the Web, but the Web also relies on HTML and browsers.